Hanwen Chen

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EDUCATION

University of California, Santa Cruz *Expected Dec 2026*
M.S. in Electrical and Computer Engineering *Santa Cruz, CA*

Shanghai University *Jun 2025*
B.E. in Electronic Information Engineering *Shanghai, China*

EXPERIENCE

Tactile Manipulation Lab, UC Santa Cruz *Sep 2025 – Present*
Graduate Researcher | Autonomous Tomato Seedling Transplantation *Santa Cruz, CA*

- Building an end-to-end Python pipeline from ZED 2i RGB-D capture to a **Franka Research 3**; demonstrated seedling acquisition, tray extraction, transport, and placement in initial end-to-end trials.
- Annotated **200+ RGB-D images** with instance masks using plant/session-level splits to prevent data leakage; fine-tuned an **Ultralytics YOLO** segmentation model, evaluating it on downstream grasp-candidate validity.
- Converted image masks into robot-frame grasp poses via confidence-based depth filtering, **RANSAC + PCA** 3D line fitting, and calibrated camera-to-base transforms; ranked candidates by depth quality, geometry, and IK feasibility.
- Implemented a fault-tolerant task **state machine** over **franky/libfranka** with pose validation, wrench-based contact detection, re-observation, bounded retries, and retreat/recovery paths for failed acquisition or extraction.
- Measured gripper capture tolerance under controlled positional offsets to derive localization accuracy requirements for the perception stack, separating holding failures from extraction resistance.

Zentek *Jun 2024 – Dec 2024*
Systems Engineering Intern *Shanghai, China*

Real-Time Conversational Digital Human GitHub

- Reduced perceived conversational latency by **60%** by streaming LLM tokens through punctuation-aware sentence segmentation (stricter first-segment rules for faster first audio) into concurrent TTS and facial-animation workers.
- Built a **FastAPI** service with WebSocket and REST endpoints orchestrating Riva ASR, a streaming LLM, GPT-SoVITS, and Audio2Face (gRPC); used asyncio for I/O and producer–consumer threads for GPU inference.
- Mapped Audio2Face output to **52 ARKit blendshapes** and GLTF morph targets; chunked audio over 10 s to fit A2F buffer limits and added a procedural fallback driven by multiband STFT features with Kalman smoothing.
- Rendered a 3D avatar at **60 FPS** with React and Three.js (React Three Fiber), syncing a queued audio/animation segment player; added timeouts, WebSocket auto-reconnect, and HTTP upload fallback for resilience.

IMU-Driven Digital Twin GitHub

- Built a **100 Hz** IMU pipeline on Jetson Orin Nano (MPU6050 over I2C), estimating roll, pitch, and yaw with Kalman filtering and removing gravity from acceleration; kept measured orientation error within **2°**.
- Streamed sensor data over HTTP into OpenUSD transforms published through Omniverse Nucleus Live collaboration sessions, achieving **sub-100 ms** measured scene latency.

TECHNICAL SKILLS

Languages & tools: Python, C++, JavaScript, Bash, Linux, Git
Backend & systems: FastAPI, asyncio, multithreading, WebSockets, REST, gRPC, Flask, FFmpeg
ML & perception: PyTorch, Ultralytics YOLO, OpenCV, scikit-image, NumPy/SciPy, Open3D, RGB-D point clouds
Speech & generative AI: streaming LLM APIs, NVIDIA Riva ASR, GPT-SoVITS TTS, NVIDIA Audio2Face
Frontend & 3D: React, Three.js, React Three Fiber, GLTF, NVIDIA Omniverse Kit, OpenUSD